

Step-Up Fabrication Document

Top Fabrication (Scale 1:1)

FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

- 1) FABRICATE PER IPC-6012A CLASS 2.
- 2) OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge_Cuts.GBR" SUFFIX.

DIMENSIONS SHOWN ON THIS DRAWING ARE FOR REFERENCE ONLY.
- 3) SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.

SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING ARE FOR REFERENCE ONLY.
- 4) SURFACE FINISH: HAL LEAD-FREE
- 5) SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR YELLOW.
- 6) SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING WHITE NON-CONDUCTIVE EPOXY INK.
- 7) ALL VIAS ARE TENTED ON BOTH SIDES UNLESS SOLDERMASK OPENED IN GERBER.
- 8) VENDOR SHOULD FOLLOW ROHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING
- 9) PCB MATERIAL REQUIREMENTS:

A. FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS.

B. Tg 170 C OR EQUIVALENT.

C. EQUIVALENT MATERIAL SHALL BE RoHS COMPLIANT, HALOGEN FREE AND APPROVED BY BUYER.
- 10) DESIGN GEOMETRY MINIMUM FEATURE SIZES:

BOARD SIZE85,000 × 100,000 mm

BOARD THICKNESS1,600 mm

TRACE WIDTH0,300 mm

TRACE TO TRACE0,020 mm

MIN. HOLE (PTH)0,250 mm

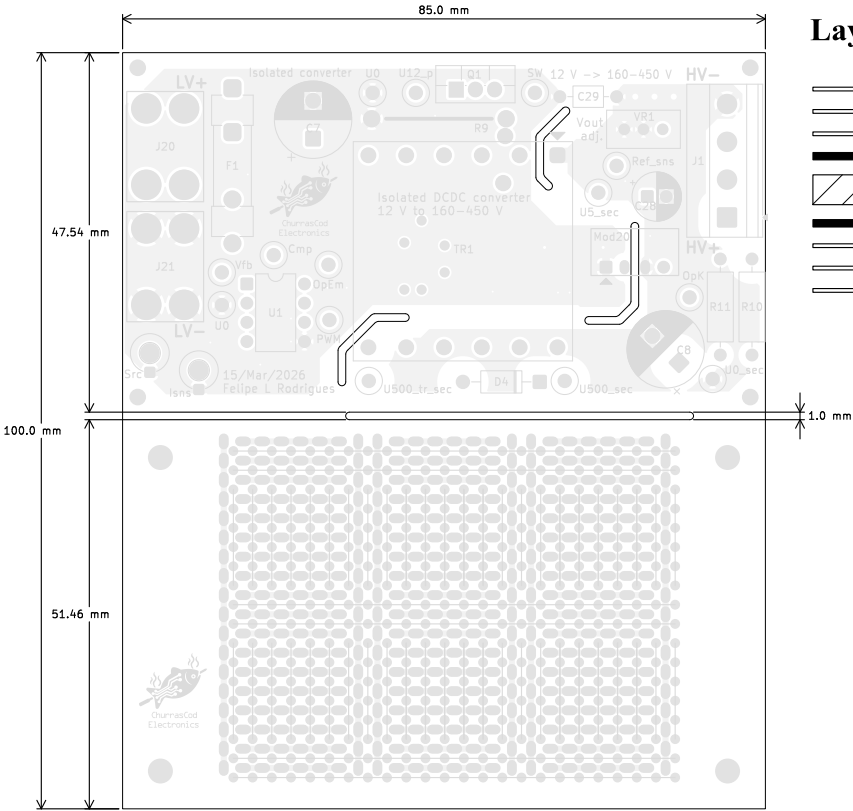
MIN. HOLE (NPTH)2,100 mm

ANNULAR RING0,150 mm

COPPER TO HOLE0,200 mm

COPPER TO EDGE0,250 mm

HOLE TO HOLE0,254 mm



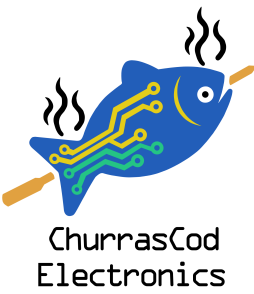
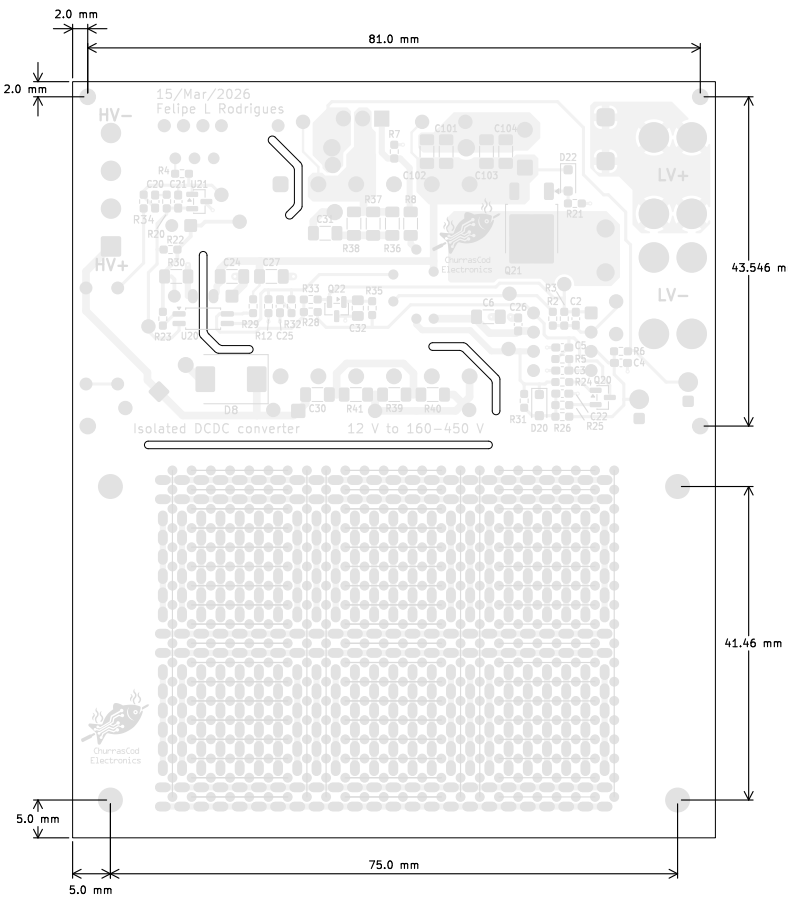
Layer Stack Legend

Material	Layer	Thickness	Dielectric	Type	Gerber
F,Paste				Paste Mask	
F,Silkscreen			Direct Printing	Legend	GBR
F.Mask		0,02 mm	Solder Resist	Solder Mask	GBR
Copper	L1 (Sig, PWR)	0,035 mm (1 oz)		Signal	GBR
Core		1,49 mm	FR4	Dielectric	
Copper	L2 (Sig, PWR)	0,035 mm (1 oz)		Signal	GBR
B.Mask		0,02 mm	Solder Resist	Solder Mask	GBR
B,Silkscreen			Direct Printing	Legend	GBR
B,Paste				Paste Mask	

Total thickness: 1,6 mm
Note: external layer thicknesses are specified after plating



Comments:	Company: NA		Variant: CHECKED	Git Hash: 8c89fcc
	Board Name: Step-Up		Project Name: Isolated step-up 12:450 V	
Sheet Title: Top Fabrication (Scale 1:1)	File Name: Isolated_12to450V.kicad_pcb	Designer: FR	Date: 2026-02-07	Revision: 2.0.0+ (Unreleased)
Sheet Path:		Reviewer: FR	Size: A4	Sheet: 1 of 10



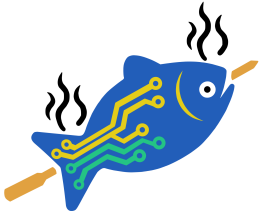
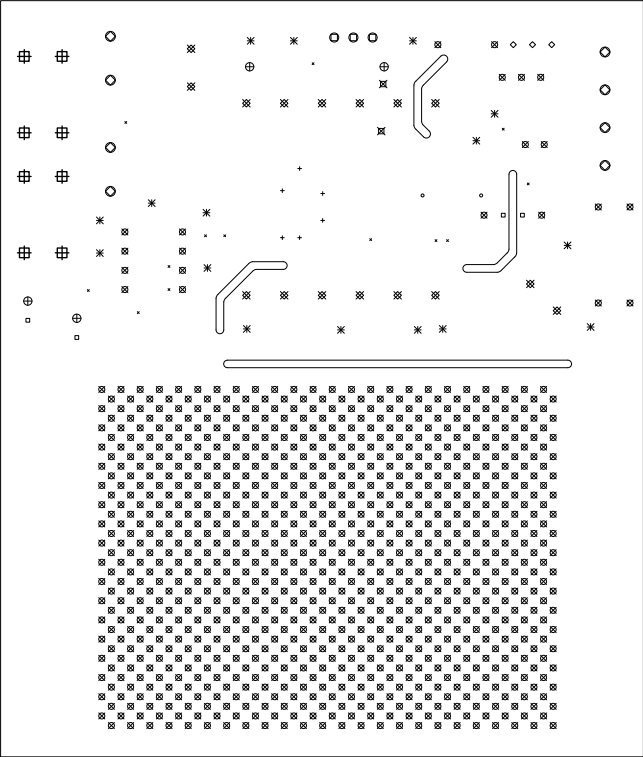
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	Board Name: Step-Up			Project Name: Isolated step-up 12:450 V		
Sheet Title: Bottom Fabrication (Scale 1:1)	File Name: Isolated_12to450V.kicad_pcb		Designer: FR		Date: 2026-02-07	Revision: 2.0.0+ (Unreleased)
Sheet Path:			Reviewer: FR		Size: A4	Sheet: 2 of 10

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Drill Drawing L1 - L4 (Scale 1:1)

Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	13	0.25mm (9,84mils)	PTH	Round	L1 (Ref) - L2 (Sig. PWR)	Via
○	2	0.40mm (15,75mils)	PTH	Round	L1 (Ref) - L2 (Sig. PWR)	Via
+	6	0.50mm (19,69mils)	PTH	Round	L1 (Ref) - L2 (Sig. PWR)	Via
□	4	0.70mm (27,56mils)	PTH	Round	L1 (Ref) - L2 (Sig. PWR)	Pad
◇	3	0.80mm (31,50mils)	PTH	Round	L1 (Ref) - L2 (Sig. PWR)	Via
⊗	885	0.80mm (31,50mils)	PTH	Round	L1 (Ref) - L2 (Sig. PWR)	Pad
✱	16	0.90mm (35,43mils)	PTH	Round	L1 (Ref) - L2 (Sig. PWR)	Pad
⊗	2	1.00mm (39,37mils)	PTH	Round	L1 (Ref) - L2 (Sig. PWR)	Via
⊗	16	1.00mm (39,37mils)	PTH	Round	L1 (Ref) - L2 (Sig. PWR)	Pad
⊕	4	1.10mm (43,31mils)	PTH	Round	L1 (Ref) - L2 (Sig. PWR)	Pad
⊕	3	1.20mm (47,24mils)	PTH	Round	L1 (Ref) - L2 (Sig. PWR)	Pad
○	8	1.30mm (51,18mils)	PTH	Round	L1 (Ref) - L2 (Sig. PWR)	Pad
⊕	8	1.80mm (70,87mils)	PTH	Round	L1 (Ref) - L2 (Sig. PWR)	Pad
Total 970						



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Comments:

Company:

NA

Variant:

CHECKED

Git Hash:

8c89fcc

Board Name:

Step-Up

Project Name:

Isolated step-up 12:450 V

Sheet Title:

Drill Drawing (L1 - L4)

File Name:

Isolated_12to450V.kicad_pcb

Designer:

FR

Date:

2026-02-07

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Size:

A4

Sheet:

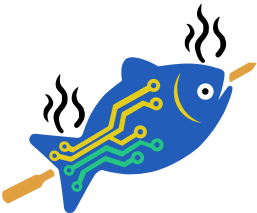
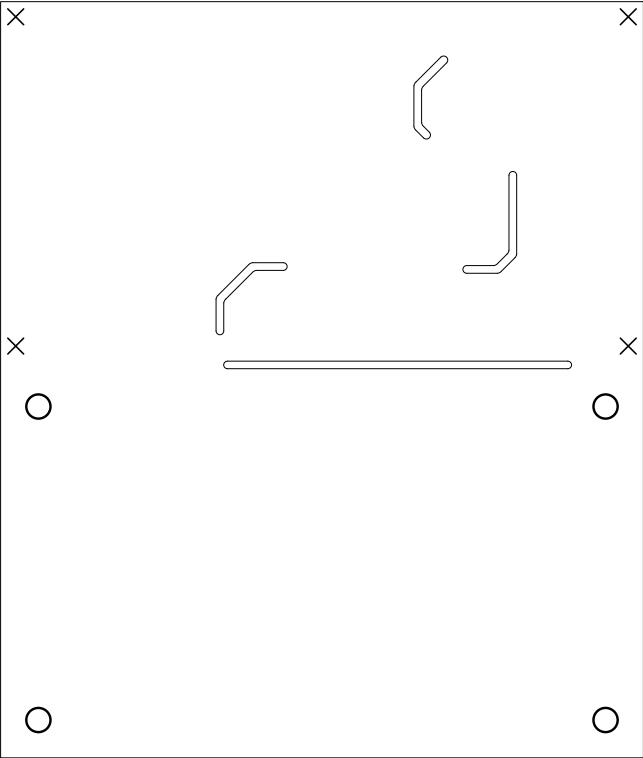
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Drill Drawing L1 - L4 (Scale 1:1)

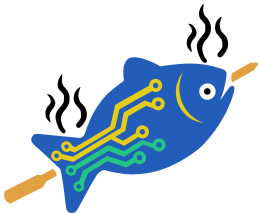
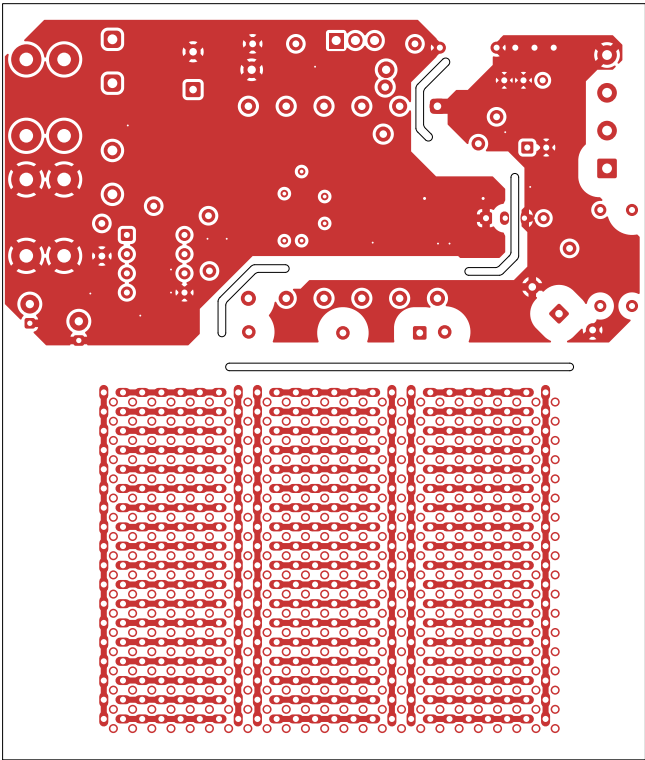
Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	4	2,10mm (82,68mils)	NPTH	Round	L1 (Ref) - L2 (Sig, PWR)	Mechanical
○	4	3,20mm (125,98mils)	NPTH	Round	L1 (Ref) - L2 (Sig, PWR)	Mechanical
Total 8						

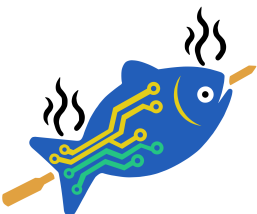


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	Board Name: Step-Up		Project Name: Isolated step-up 12:450 V	
Sheet Title: Drill Drawing (L1 - L4)	File Name: Isolated_12to450V.kicad_pcb	Designer: FR	Date: 2026-02-07	Revision: 2.0.0+ (Unreleased)
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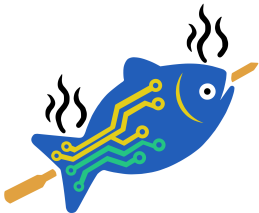
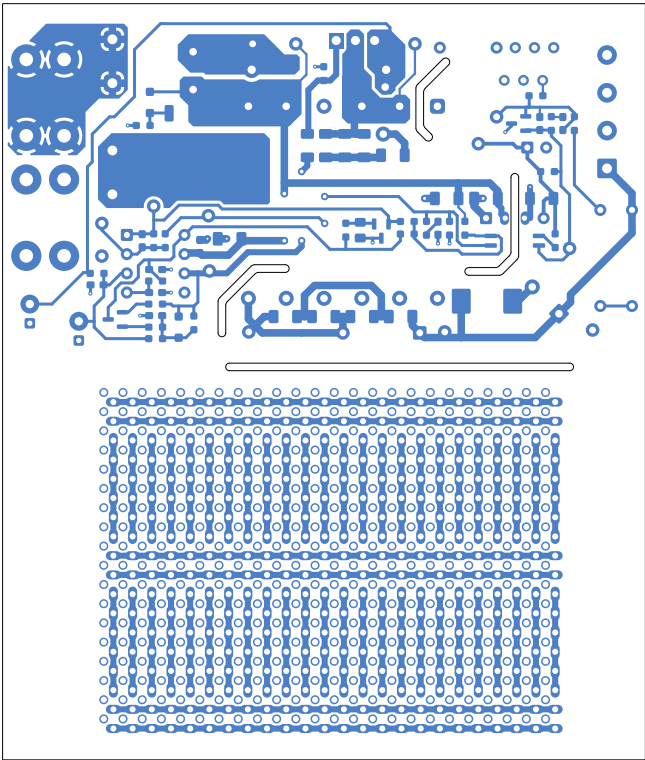


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			Step-Up			Isolated step-up 12:450 V			
	Sheet Title:		File Name:		Designer:		Date:		Revision:
F.Cu (Scale 1:1)		Isolated_12to450V.kicad_pcb		FR		2026-02-07		2.0.0+ (Unreleased)	
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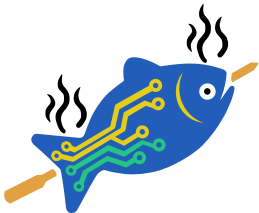
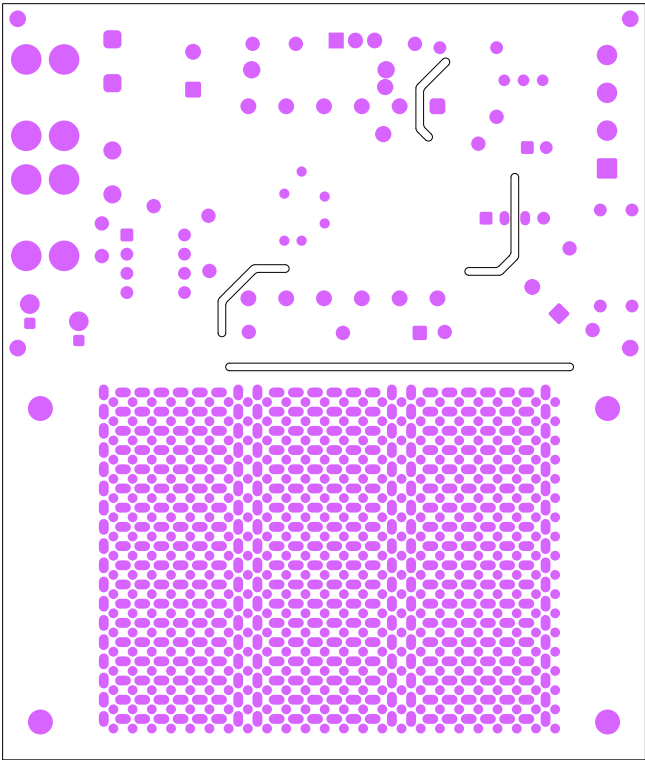
Step-Up Fabrication Document

B.Cu (Scale 1:1)

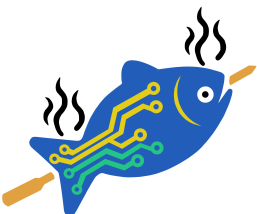


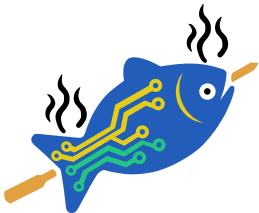
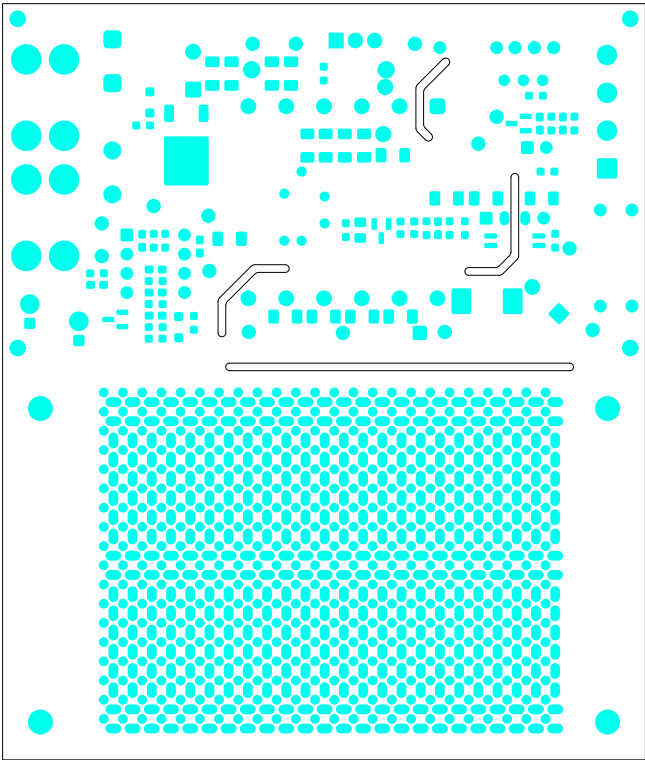
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Sheet Path:		Reviewer: FR	Size: A4	Sheet: 6 of 10

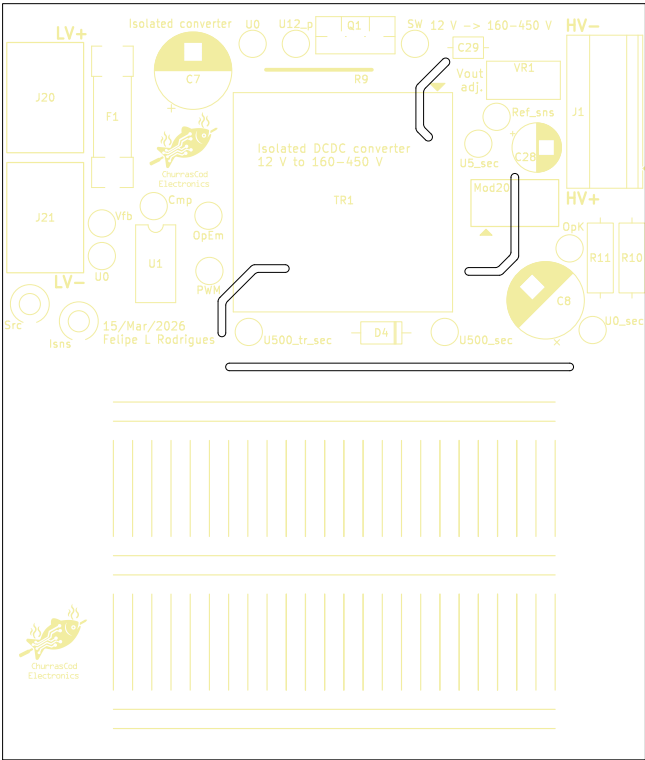


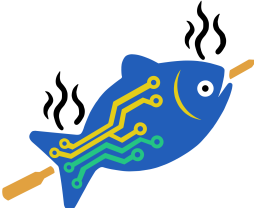
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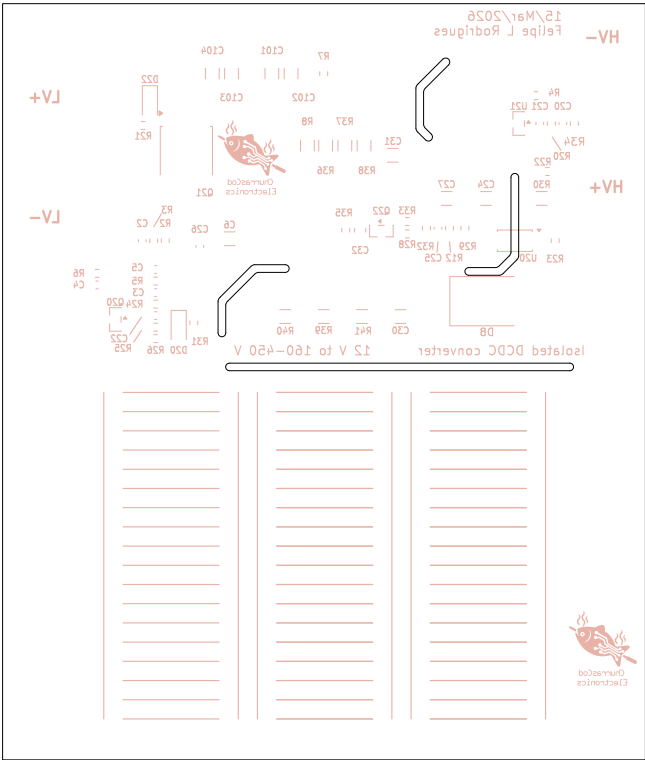
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			Step-Up			Isolated step-up 12:450 V			
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F.Mask (Scale 1:1)		Isolated_12to450V.kicad_pcb		FR		2026-02-07		2.0.0+ (Unreleased)	
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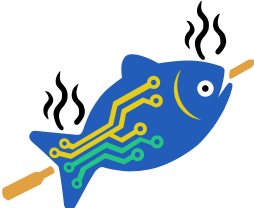


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B.Mask (Scale 1:1)		Isolated_12to450V.kicad_pcb		FR		2026-02-07		2.0.0+ (Unreleased)		
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	Sheet Title: F.Silkscreen (Scale 1:1)		Board Name: Step-Up		Project Name: Isolated step-up 12:450 V	
	Sheet Path:		File Name: Isolated_12to450V.kicad_pcb	Designer: FR	Date: 2026-02-07	Revision: 2.0.0+ (Unreleased)
			Reviewer: FR	Size: A4	Sheet: 9 of 10	



 ChurrasCod Electronics	Comments:		Company: NA		Variant: CHECKED	Git Hash: 8c89fcc
	Sheet Title: B.Silkscreen (Scale 1:1)		Board Name: Step-Up		Project Name: Isolated step-up 12:450 V	
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			Reviewer: FR	Size: A4	Sheet: 10 of 10	